Gossip Principle Analysis

Imagine the network as a population of $n+1$ individuals:

- y (Infected): Nodes that have received the information.
- x (Uninfected): Nodes that have not yet received the information
- $x + y = n + 1$: The total population remains constant.
- β (Contact Rate): The probabillity that two nodes meet and the message is transmitted.

Init $t = 0$: Only 1 node knows the news ($y_0 = 1$), and n nodes are unaware ($x_0 = n$).

We wan to know

$$\frac{dx}{dt} = -\beta xy$$

Why $-\beta xy$?

First, any pair (x, y) that meets will result in a message transmission. There are totally xy possible pairs and each pair has a β chance to transmit the message; the negative sign indicates that x should decrease over time.

$$\Rightarrow y = \frac{n+1}{1 + n \cdot e^{-\beta(n+1)t}} \quad \text{(S curve)}$$

However, In a Gossip protocal, we use "active selection" rather than "random collision":

Each node randomly pick b targets every round (aka. Fan-out). Since there are n potential targets, the contact rate is $\beta \approx \frac{b}{n}$. According to the most powerful feature of the Gossip protocol, the time required for informantion to disseminate across the entire network is $O(\log n)$, we set time as $t = c \log n$ (where c is a constant).

$$\Rightarrow y \approx \frac{n+1}{1 + n \cdot n^{-bc}} = \frac{n+1}{1 + n^{1-bc}}$$

Key Takeaways:

1. Efficiency: Reaches the entire network in $O(\log n)$ rounds.
2. Scalability: Each node carries a constant load of b messages per round, regardless of network size.